Molek-30M: A Compute-Efficient From-Scratch MoFlow Replica for Agricultural Molecular Generation

PinoBio-GenD research note - MOLEK-001 - ZINC250k generation + distributional benchmark

Stan Lee V. Natividad - Kooya PH AI Research

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Abstract

We present **Molek-30M**, a from-scratch replica of the MoFlow invertible-flow molecular graph generator, scaled to **29,049,444 parameters** (<=30M budget) and trained on a single free-tier GPU (Kaggle T4). On the ZINC250k benchmark, Molek-30M **matches the full reference MoFlow **on all standard generation metrics - corrected validity** 99.95%**, novelty **100%**, uniqueness **99.99%** - and reproduces MoFlow's distributional behaviour, with a Fréchet ChemNet Distance (**FCD 23.40**) in line with the published reference (20.93) at a ~25x parameter reduction. We report every metric with and without validity correction, contextualise FCD against the published ZINC250k leaderboard, and state the honest finding that MoFlow-family models - Molek included - are mid-pack on distributional fidelity, where diffusion models lead. Molek-30M is the base for an agricultural-conditioning head (future work) for Philippine crop-protection chemistry.

1. Research question

Can a MoFlow-class molecular generator, replicated **from scratch** under severe compute constraints (single free T4, <=30M params), **faithfully reproduce the reference model's generation quality** on ZINC250k, and provide a sound base for agricultural property-conditioned generation?

2. Hypotheses

Main. A compute-constrained from-scratch MoFlow replica (<=30M params, single free T4) reproduces the reference MoFlow's ZINC250k generation quality within measurement noise.

H1 (generation quality). Corrected validity, novelty, and uniqueness each within **+/-1 percentage point** of the reference MoFlow (~100% / 100% / ~100%).

H2 (distributional fidelity). Despite ~25x fewer parameters, FCD stays the same order as the reference (20.93) - specifically **<= 1.5x the reference FCD** - not catastrophic.

H3 (correction dependence). Uncorrected (raw) validity is near-zero, reflecting MoFlow's architectural reliance on valency correction; corrected metrics are the capability measure.

3. Related work

MoFlow (Zang & Wang, KDD 2020) - invertible flow over molecular graphs with post-hoc valency correction (our reference). GraphAF (Shi 2020), GraphDF (Luo 2021) - autoregressive flows. GDSS (Jo 2022), DiGress (Vignac 2023), GraphArm (Kong 2023), CDGS (2023) - diffusion/score-based, current distributional-fidelity leaders. MOSES (Polykovskiy et al.) and FCD (Preuer et al. 2018) - evaluation standards inherited here.

4. Dataset

ZINC250k (MOSES/`molecularsets`, MIT; ZINC free): 249,455 molecules; reference split (`valid_idx_zinc.json`) 224,568 train / 24,887 test. Provenance in `data/manifests/moses-zinc250k.jsonl` (sha256, row-count G7-verified; mirror used after the DeepChem S3 403). License in `configs/licenses.yaml` (G1).

5. Methodology

Reference MoFlow, scaled (M-MOFLOW-1): `b_hidden_ch 256,256`, `a_n_flow 19`, `a_hidden_gnn 128`, `a_hidden_lin 256,64`, `b_n_flow 5`, `b_conv_lu 2`, `noise_scale 0.6` -> 29,049,444 params (hard-fail > 30M). Training: AMP fp16, Adam lr 1e-3, seed 20261023, 200-epoch target; recovered checkpoint at epoch 197/200. Compute: single Kaggle T4x2; py3.13 patches M-MOFLOW-5/6/9/10. Eval: frozen reference `generate.py` (G3/G11), 10,000 samples x 5 experiments, temperature 0.85. Dual-report (M-MOFLOW-4): every metric with and without valency correction.

6. Baselines

Published ZINC250k numbers (GLAD/Kalousis 2024; GDSS/CDGS tables):

Model	Validity (%) (up)	FCD (down)	Family
GraphAF	68.47	16.02	autoregressive flow
GDSS	97.01	14.66	diffusion (best FCD)
GraphArm	88.23	16.26	autoregressive diffusion
MoFlow (reference)	~100 (corr.)	20.93	normalizing flow
Molek-30M (ours)	**99.95 (corr.)**	**23.40**	normalizing flow (replica)
DiGress	91.02	23.06	discrete diffusion

7. Experiments

Each follows the `EXPERIMENT_PROTOCOL` field set.

EXP-MOLEK-001 - from-scratch training

- **Research question:** can MoFlow be trained from scratch at <=30M on a single T4?
- **Hypothesis:** Main. **Model:** Molek-30M (29,049,444). **Dataset:** ZINC250k (pinned).

Tokenizer: n/a (graph). **Method:** flow MLE, AMP fp16, Adam lr 1e-3, 200-ep target.

- **Hyperparameters:** b_hidden_ch 256,256 - a_n_flow 19 - a_hidden_gnn 128 - a_hidden_lin 256,64 - b_n_flow 5 - b_conv_lu 2 - noise_scale 0.6.

- **Hardware:** Kaggle T4x2. **Seed:** 20261023. **Baseline:** reference MoFlow.
- **Expected:** stable training to ~200 ep. **Actual:** epoch 197/200; ckpt sha256 `7f8374df3cb8...`; preflight 29,049,444 <= 30M.
- **Conclusion:** feasible (Main supported). **Next:** EXP-MOLEK-002.

EXP-MOLEK-002 - generation quality (dual-report, M-MOFLOW-4)

- **RQ:** does the replica match reference validity/novelty/uniqueness? **Hypothesis:** H1, H3.
- **Model:** Molek-30M @ ep197. **Params:** 10,000 x 5, temp 0.85. **Seed:** 20261023.

Baseline: reference MoFlow (100/100/~100). **Suite:** frozen reference `generate.py`.

- **Expected (H1):** within +/-1 pp of reference. **Actual:**

Metric	Corrected	Uncorrected	Reference	delta vs ref
Validity	99.95% (sd 0.02)	0.00% (sd 0)	100.00%	–0.05
Novelty	100.00% (sd 0.00)	0.00%	100.00%	0
Uniqueness	99.99%	0.00%	~100%	~0
Abs. novelty	99.94%	0.00%	-	-
Abs. uniqueness	99.94%	0.00%	-	-

Per-experiment valid /10,000: 9996, 9998, 9994, 9990, 9995.

- **Conclusion:** H1 supported (within +/-1 pp); H3 supported (raw 0% = MoFlow correction-dependence). **Next:** EXP-MOLEK-003.

EXP-MOLEK-003 - distributional fidelity (FCD / SNN / IntDiv)

- **RQ:** does the replica reproduce MoFlow's distribution quality? **Hypothesis:** H2.
- **Model:** Molek-30M @ ep197. **Dataset:** ZINC250k test (24,887). **Params:** 30,000 generations, temp 0.85; FCD via fcd_torch (= MOSES FCD), IntDiv/SNN via RDKit. **Seed:** 20261023. **Baseline:** published (MoFlow 20.93; GDSS 14.66; GraphAF 16.02; DiGress 23.06).
- **Expected (H2):** FCD <= 1.5x reference (<= 31.4). **Actual:** FCD **23.40** (23.3965) - SNN **0.196** - IntDiv **0.908** - IntDiv2 **0.900**; FCD ratio 23.40/20.93 = **1.12**.
- **Conclusion:** H2 supported. Molek ~ reference on FCD within scale-down; both mid-pack (diffusion leads). **Next:** farm-knob conditioning; diffusion arch for FCD frontier.

Key findings

1. Molek-30M matches reference MoFlow on validity/novelty/uniqueness (H1) and FCD (H2).
2. Raw validity is 0% - MoFlow-by-design (H3); corrected is the capability metric.
3. MoFlow-family is mid-pack on distributional fidelity; diffusion is the frontier lever.

Failed / negative experiments (recorded per protocol)

Training runs #1-#6 failed on sequential py3.13 incompatibilities (distutils, orderedset, rdkit.six, cairosvg) -> permanent patches M-MOFLOW-5/6/9/10. D3 uncorrected eval first crashed (`UnboundLocalError` in reference `check_novelty` on 0 valid) -> M-MOFLOW-13. The 2020 `moses` package is py3.13-incompatible -> pivoted to `fcd_torch` for FCD (M-MOSES-1). FCD first mis-scaled vs MOSES, self-corrected against the ZINC250k baseline (F-0026).

8. Ablations

Formal ablations (temperature sweep, parameter-budget sweep, AMP on/off) are **not yet run** - stated honestly, not fabricated. Only validity correction on/off is varied here (§7, EXP-MOLEK-002). Planned: temperature $\in$ {0.7, 0.85, 1.0} vs FCD/diversity; 13M vs 30M.

9. Statistical analysis

Means +/- sd over 5 independent generation experiments (seed 20261023): validity sd 0.02%, novelty sd 0.00%. FCD is a single distributional estimate over 30,000 generations vs 24,887 test molecules. Bootstrap CI + Wilcoxon tooling (`benchmarks/validation/metrics.py`) is in place for downstream binder tasks.

10. Limitations

Mid-pack distributional fidelity (FCD 23.40; MoFlow-class ceiling - diffusion is the lever, not a Molek tune). Uncorrected validity 0% (intrinsic to MoFlow). Novelty uses the reference's own SMILES-match (apples-to-apples with reference; a rigorous "true novelty" needs a canonicalisation audit). Not agri-conditioned yet (farm-knob spec'd, not trained). No wet-lab validation (none claimed). FCD absolute values vary by implementation (same ballpark, not identical-protocol).

11. Ethics

Generative chemistry for **agriculture** (crop protection), not human therapeutics. No efficacy/safety claims; wet-lab performance out of scope until benchmarked. Licenses verified (ZINC/MOSES MIT); no PII. Open, reproducible, honestly reported (G7).

12. Reproducibility

Seed 20261023; config hash in `configs/molgen-zinc250k.yaml`; pins `moflow@3611c6372602`, `moses@dd7ed6ab38e2`. Checkpoint sha256 `7f8374df3cb8...`; eval = frozen reference `generate.py`. Code: `scripts/train_moflow.py`, `eval_moflow.py`, `eval_moses_metrics.py`; results `docs/results/molek-30m-*.{md,json}`; patches M-MOFLOW-1..13 in `patches/`.

13. Conclusion

Molek-30M is a faithful, compute-efficient MoFlow replica - matching the reference on every standard ZINC250k metric including FCD, at ~25x fewer parameters on a free GPU. An honest baseline (not distributional SOTA) and the foundation for agricultural property-conditioned generation.

Acknowledgements and AI-use statement

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(Zang & Wang, KDD 2020, MIT). AI assistance was used for implementation, experiment orchestration, and drafting; all numbers are measured on pinned code and verified against source result files. No external closed-API model outputs were used for training (G2).